Practice workbook · Link a nutrient to its function, source and deficiency disease

Functions and food sources of named nutrients and the deficiency diseases scurvy, anaemia, vitamin A blindness and kwashiorkor.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-NUTRIENT-FUNCTION-DEFICIENCY

1. Question 2 · part b · 2 marks

June 2019 Paper 1R

Worked-example retry · Attempt part b only.

2 Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

(2)

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

(2)

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1. Question 2 · part b · 2 marks

June 2019 Paper 1R

Worked-example retry · Attempt part b only.

(c) The vitamin C content of fruit juices varies.

(i) Design an investigation to compare the vitamin C content of fresh fruit juice and long-life fruit juice.

(5)

(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(Total for Question 2 = 11 marks)

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1. Question 2 · part b · 2 marks · Working and answer

Working and answer

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2. Question 4 · part c · 1 mark

June 2019 Paper 2

Further practice · Attempt part c only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

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2. Question 4 · part c · 1 mark

June 2019 Paper 2

Further practice · Attempt part c only.

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(d) Describe a method that Goldberger could use to show that the symptoms of his patients were caused by a lack of vitamin B in their diet.

(3)

(e) Scurvy is another deficiency disease.

Scurvy affects the production of a protein called collagen.

Collagen is a component of many body tissues including cartilage.

Explain why a person with scurvy often experiences pain when walking.

(2)

(Total for Question 4 = 9 marks)

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Working and answer

3. Question 1 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

1 (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.



(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

(1)

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

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3. Question 1 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) The diagram shows the nutritional label on a package containing yogurt.

Nutrition Facts

Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

Amount/Serving	% Daily Value*	Amount/Serving	% Daily Value*
Total Fat	a%	Sodium	2%
Saturated Fat	0%	Total Carb.	6%
Trans Fat		Sugars	
Cholesterol	1%	Protein	22%

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

(1)

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

(1)

- ☒ **A** it contains vitamins C and D
- ☒ **B** it contains less than 5 mg of cholesterol
- ☒ **C** it does not contain energy from carbohydrates
- ☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

(3)

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(Total for Question 1 = 6 marks)

3. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 4 · part c.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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4. Question 4 · part c.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.i only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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4. Question 4 · part c.i · 1 mark · Working and answer

Working and answer

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5. Question 4 · part c.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.ii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

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In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

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5. Question 4 · part c.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.ii only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

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(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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5. Question 4 · part c.ii · 1 mark · Working and answer

Working and answer

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6. Question 4 · part a · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a only.

4 (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

(2)

fingers	carbohydrate	bread	ankles	protein	fat

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

Year	Number of cases of kwashiorkor
2010	32
2011	21
2012	23
2013	19

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6. Question 4 · part a · 2 marks · Working and answer

Working and answer

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7. Question 4 · part b.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

4 (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

(2)

fingers	carbohydrate	bread	ankles	protein	fat

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

Year	Number of cases of kwashiorkor
2010	32
2011	21
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2013	19

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7. Question 4 · part b.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

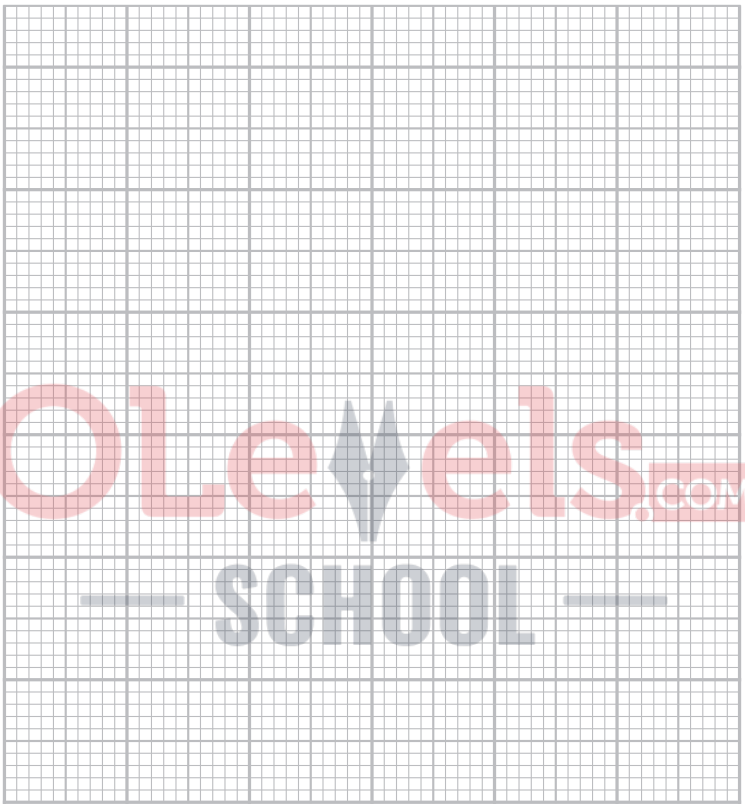
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(i) Plot a bar chart of the data on the grid.

(3)



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(ii) Suggest a reason for the difference in the number of recorded cases of kwashiorkor in 2010 compared with 2013.

(1)

.....

.....

(iii) In a hospital, 95 out of a total of 1978 children with malnutrition were found to have kwashiorkor.

Calculate the percentage of children with kwashiorkor.

(2)

percentage of children = ... %

(Total for Question 4 = 8 marks)

7. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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8. Question 5 · part a.i · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.i only.

5 (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

(1)

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

(4)

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8. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

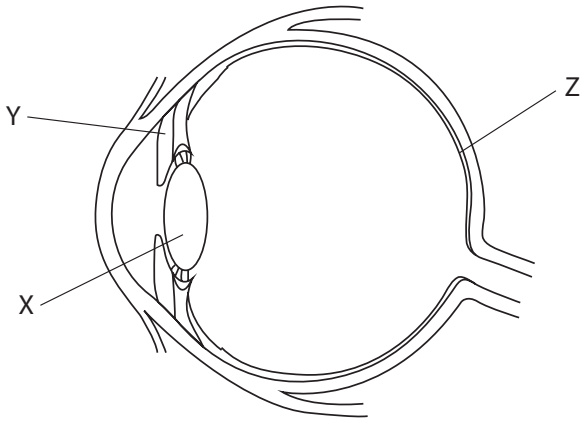
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9. Question 1 · part b.iv · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.iv only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
	transmits electrical impulses to the brain
structure X	refracts light rays
	protects the eye from pathogens
structure Y	
	detects light rays
structure Z	controls the shape of the lens
	controls the size of the pupil

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9. Question 1 · part b.iv · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.iv only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

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(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

9. Question 1 · part b.iv · 1 mark · Working and answer

Working and answer

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10. Question 4 · part d · 5 marks

June 2023 Paper 1R

Further practice · Attempt part d only.

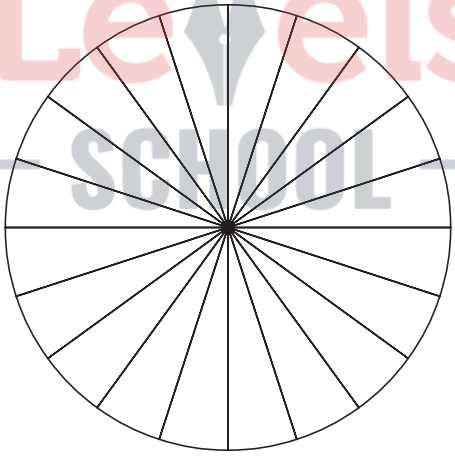
4 The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

Component	Percentage (%) composition of diet	
	Area A	Area B
cereals	75.0	15.0
fruit and vegetables	15.0	37.5
eggs, fish and meat	2.5	27.5
milk, cheese and butter	7.5	20.0

(a) Complete the pie chart to show the diet of the girl in Area B.

(4)



(b) Explain what is meant by a balanced diet.

(2)

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10. Question 4 · part d · 5 marks

June 2023 Paper 1R

Further practice · Attempt part d only.

(c) Calculate the ratio of the percentage composition of cereals in the diet of the girl in Area A compared with the diet of the girl in Area B.

Give your answer in the form n : 1

(2)

ratio = ...

(d) Discuss the effects that the percentage composition given for eggs, fish and meat in the diet may have on the growth of the two girls.

(5)

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(Total for Question 4 = 13 marks)

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10. Question 4 · part d · 5 marks · Working and answer

Working and answer

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11. Question 3 · part a.iv · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iv only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
25 to 50 years old	12 200	63	0.9	60	10
FEMALES					
15 to 18 years old	9200	46	0.7	50	15
25 to 50 years old	9200	50	0.7	60	15

(i) Describe what is meant by a balanced diet.

(2)

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(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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11. Question 3 · part a.iv · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iv only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

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.....

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.....

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

(1)

.....

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

(4)

extra mass = ... mg

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11. Question 3 · part a.iv · 1 mark · Working and answer

Working and answer

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12. Question 3 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
25 to 50 years old	12 200	63	0.9	60	10
FEMALES					
15 to 18 years old	9200	46	0.7	50	15
25 to 50 years old	9200	50	0.7	60	15

(i) Describe what is meant by a balanced diet.

(2)

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(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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12. Question 3 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

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.....

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(4)

extra mass = ... mg

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12. Question 3 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

(b) In the 19th century more than two million sailors on long ocean voyages died from a nutritional disease called scurvy.

(i) Give a symptom of scurvy.

(1)

.....

.....

(ii) Suggest why scurvy is an uncommon nutritional disease in the modern world.

(1)

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(Total for Question 3 = 14 marks)

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Working and answer

13. Question 3 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
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15 to 18 years old	9200	46	0.7	50	15
25 to 50 years old	9200	50	0.7	60	15

(i) Describe what is meant by a balanced diet.

(2)

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(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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13. Question 3 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

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.....

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(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

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Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

(4)

extra mass = ... mg

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13. Question 3 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

(b) In the 19th century more than two million sailors on long ocean voyages died from a nutritional disease called scurvy.

(i) Give a symptom of scurvy.

(1)

.....

.....

(ii) Suggest why scurvy is an uncommon nutritional disease in the modern world.

(1)

.....

.....

(Total for Question 3 = 14 marks)

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Working and answer